Franklin Liu

Chicago, IL | (312) 539-2390 | franklinliu@uchicago.edu | github.com/frank22lin |

EDUCATION

The University of Chicago Chicago, IL
Master of Science in Financial Mathematics **(GPA: 4.0/4.0)** September 2025 – December 2026
Concentration in Machine Learning and Statistics

Technical University of Berlin Berlin, Germany
Bachelor of Science in Mathematics, Minor in Computer Science **(GPA: 3.8/4.0)** October 2024
Focus: Stochastic Processes, Measure Theory and Mathematical Machine Learning

University of California, Los Angeles Los Angeles, CA
Exchange student for Mathematics **(GPA: 4.0/4.0)** September 2023 – April 2024
Awarded merit-based scholarship and joined Prof. Guido Montúfar's machine learning research group.
Research project on deriving asymptotic bounds on sequences from singularities with Prof. Terence Tao.

WORK EXPERIENCE

Quantitative Research Intern | Python, Git Berlin, Germany
Ultramarin September 2024 – May 2025

- Conducted alpha research for European equities identified exploitable correlation patterns. Improved sharpe ratio of existing strategies by **80%**, which resulted in the deployment of live signal
- Developed tools in Python that was adopted by research and portfolio management team, including automated performance tracking of strategies, backtester using order book data for realistic strategy implementation and cross-validation framework. Improved data pipeline with speedup of **150%**

Undergraduate Teaching Assistant for Probability Theory Berlin, Germany
Technical University of Berlin April – December 2023

- Held weekly exercise classes for upper-class undergraduate math students in probability theory; responsible for conducting office hours, grading homework and final exam

PERSONAL PROJECTS

Arbitrage Trading & MM on Polymarket | Python June 2025 – Current

- Noticed price inefficiencies on Polymarket between logically equivalent markets on Polymarket. Built and deployed automated trading system hosted on AWS with real-time monitoring and automatic execution, achieving **80%** return in 4 months. Working on market-making system using asynchronous WebSockets.

RESEARCH EXPERIENCE

Research Assistant at University of Chicago Booth October 2025 – Present

- **Supervised by** Prof. Dacheng Xiu
- Reconstructed millisecond-level the full-depth Shenzhen Stock Exchange limit order book from raw data
- Developed modular and config-based simulation environment with performance tracking using Polars, where multiple agents can interact in a shared order book, to research optimal execution problems

Thesis: *Modelling Limit Order Books using Diffusion Model* April – October 2024

- **Supervised by** Prof. Peter Friz, Prof. Klaus-Robert Müller
- Modelled order books in continuous time as a point process and novel application of data-driven conditional diffusion model to model snapshots; successfully replicated stylized facts (Link to thesis)
- Awarded best-possible grade and presented thesis at UCLA Machine-learning group

SKILLS AND EXTRACURRICULAR

Computing: Python (Pandas, Polars, Scikit-learn, Scipy, Pydantic), Git, AWS, LaTeX
Competitions: Best data-analysis for Midwest Blockchain Conference Research Competition ($2500 award), CME Trading Competition (2nd Place out of 600+, $1500 award), IMC Prosperity (Top 1% globally)
Clubs: Blockchain Chicago